

Standard mode 14:11

$sublista([], -)$
 $sublista([H|T], [H|L]) :- sublista(T, L).$
 $sublista([H|T], [_|L]) :- sublista([H|T], L).$
 $sublista(L, LS) :- setof(S, sublista(S, L), LS).$
 $invel([], [])$
 $invel([X|Y], [X|U]) :- invel(Y, U).$
 $totala(R, A, B) :- not((member(X, A), not(member(X, B), not(member(X, R))))).$
 $surjectiva(R, A, B) :- invel(R, A), totala(B, B, A).$
 $permutare([], [])$
 $permutare([H|T], P) :- permutare(T, Q), sterge(H, P, Q).$
 $permutabile(L, LP) :- setof(P, permutare(L, P), LP).$
 $egalmult(L, M) :- permutare(L, M)$
 $sterge([], -) :- fail.$
 $sterge([H|T], T)$
 $sterge(X, [H|T], [H|L]) :- sterge(X, T, L).$
 $?- sterge(a, [a, 1, a, 3, 3, a], L).$
 $?- sterge(a, Diferlistay[1, 3, 3]).$
 $?- permutare([1, 3, 3], P).$
 $?- permutare(L, [1, 3, 3], P).$
 $sterge(H, P, Q)$

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$submult(S, M) :- permutare(M, P)$
 $sublista(S, P)$
 $?- submult([1, 3, 0], [0, 1, 3, 3, 4, 5]).$
 $subset(R, B) :- not((member(X, B), not(member(X, R)))).$
 $subset(R) :- not((member(X, B), not(member(X, R)))).$
 $member(X, B, R, A \setminus = X).$
 $reunire(L, M, R) :- setof(X, (member(X, L), member(X, M)), R), !.$
 $reunire([], []).$
 $intersectie(L, M, I) :- setof(X, (member(X, L), member(X, M)), I), !.$
 $intersectie([], []).$
 $diferenta(L, M, D) :- setof(X, (member(X, L), not(member(X, M))), D), !.$
 $diferenta([], []).$
 $xor(P, Q) :- P, not(Q); not(P), Q.$
 $diferentasimetrica(L, M, D) :- setof(X, xor(member(X, L), member(X, M)), D), !.$
 $diferentasimetrica(L, -, []).$

Standard mode 14:10

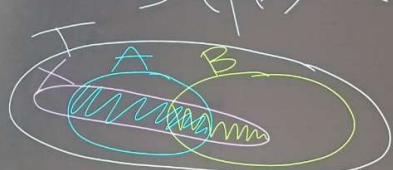
$\text{in}(A, \text{In}) :- \text{setof}(X^1 (\text{member}(X, A) \wedge \text{member}(X, Y, R)), \text{In}), !$
 $\text{in}(_, _)$
 $\text{prein}(R, B, \text{PreIn}) :- \text{setof}(X^1 (\text{member}(X, B) \wedge \text{member}(X, Y, R)), \text{PreIn}), !$
 $\text{prein}(_, _)$
 $\text{img}(_, _)$
 $\text{img}([X, Y] | T, A, [X] | L) :- \text{member}(X, A), !$
 $\text{img}([X, Y] | T, A, L) :- \text{img}(T, A, L)$
 $\text{preimg}(_, _)$
 $\text{preimg}([X, Y] | T, B, [X] | L) :- \text{member}(Y, B), !$
 $\text{preimg}([X, Y] | T, B, L) :- \text{preimg}(T, B, L)$

$R \subseteq M \times N$
 $f: M \rightarrow N$
 $f^{-1}(B)$
 $f(A)$
 $\{a, f(a) \mid a \in M\}$

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Test, $A, B \in \mathcal{P}(T)$
 Ex: $f: \mathcal{P}(T) \rightarrow \mathcal{P}(A) \times \mathcal{P}(B)$
 $(\forall X \in \mathcal{P}(T)) (f(X) = (X \cap A, X \cap B))$

$(\forall M \in \mathcal{P}(T)) (M = T, M)$

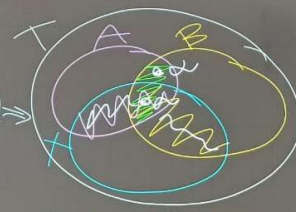


(1) $f \rightarrow \text{img} \Leftrightarrow A \cup B = T$
 (2) $f \rightarrow \text{img} \Leftrightarrow A \cap B = \emptyset$
 (3) $f \rightarrow \text{img} \Leftrightarrow A = \overline{B} \Leftrightarrow B = \overline{A}$

$A \cup B = T \Leftrightarrow A \supseteq \overline{B} \Leftrightarrow B \supseteq \overline{A}$
 $A \cap B = \emptyset \Leftrightarrow A \subseteq \overline{B} \Leftrightarrow B \subseteq \overline{A}$

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$\forall x \in T, A \cap B = \emptyset$
 Fie $(x, y) \in f(A) \times f(B) \Rightarrow$



$f(x) = (x \cap A, x \cap B)$
 $\Rightarrow X \cup Y \subseteq A \cup B \subseteq T$
 $f(X \cup Y) = ((X \cup Y) \cap A, (X \cup Y) \cap B) = (X, Y) \Rightarrow f \rightarrow \text{surj.}$
 $(X \cap A) \cup (Y \cap A) = X \cap A \cup Y \cap A = (X \cup Y) \cap A$
 $(X \cap B) \cup (Y \cap B) = X \cap B \cup Y \cap B = (X \cup Y) \cap B$
 $\Rightarrow A \cap B = \emptyset$

$\Rightarrow \neg$: $f \rightarrow \text{surj.}$ p.als. $A \cap B \neq \emptyset \Leftrightarrow \exists x \in A \cap B \Rightarrow x \cap A = x \cap B = \{x\} \Rightarrow$
 $(\exists x \in f(T)) (f(x) = (\emptyset, \{x\})) \Rightarrow \exists x \cap B = \{x\} \Rightarrow$
 $\Rightarrow (x \cap A, x \cap B)$

$A \cap B = \emptyset \Rightarrow x \cap A = \emptyset$
 $\Rightarrow x \cap B = \{x\}$

Standard mode 14:08

$\text{echiv}(P, Q) :- \text{not}(P), Q.$
 $\text{echiv}(P, Q) :- \text{imply}(P, Q), \text{imply}(Q, P)$
 $\text{fct} F(F, T) :- \text{sublista}(A, T), \text{sublista}(B, T), \text{functia}(F, T, A, B)$
 $\text{functia}(F, T, A, B) :- \text{setof}((X, X \cap A, X \cap B), (\text{sublista}(X, T), \text{intersect}(X, A, X \cap A), \text{intersect}(X, B, X \cap B)), F).$
 $\text{echiv2}(T) :- \text{not}((\text{sublista}(A, T), \text{sublista}(B, T), \text{functia}(F, T, A, B), \text{not}(\text{echiv}(\text{surject}(F), \text{reunune}(A, B, U), \text{egalmult}(T, U))))))$

$\text{echiv2}(T) :- \text{not}((\text{sublista}(A, T), \text{sublista}(B, T), \text{functia}(F, T, A, B), \text{not}(\text{echiv}(\text{surject}(F), \text{intersect}(A, B, U))))))$

Nu putem folosi variantele de mai sus ale predicatelor care testeaza injectivitatea si surjectivitatea, pentru ca acestea functioneaza doar pentru relatii intre multimi date ca liste de constante.

Iar predicatul pentru calculul intersectiei nu poate fi folosit astfel, cu constanta pe al treilea argument; trebuie calculata intersectia, cu variabila pe al treilea argument, apoi unificat rezultatul calculului cu constanta lista vida.

A se vedea varianta corecta in Seminarul 4.